

VILHELMINA, Sweden

WMO: 022450

Lat: 64.580N

Lon: 16.839E

Elev: 1145

StdP: 14.10

Time Zone: 1.00 (EUC)

Period: 96-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-25.3	-19.8	-32.6	0.9	-25.4	-26.1	1.3	-19.8	21.7	33.3	18.5	31.6	0.7	110	0.697

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.9	76.4	60.2	72.7	58.9	69.2	57.4	62.8	71.8	60.9	69.2	59.0	66.7	6.9	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
59.6	79.5	65.7	57.5	73.7	63.3	55.6	68.7	61.9	28.9	72.0	27.5	69.3	26.2	66.6	70.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
18.3	15.5	13.4	-31.4	79.7	4.7	3.9	-34.7	82.5	-37.5	84.8	-40.1	86.9	-43.6	89.8		
			-31.6	64.9	4.7	2.3	-34.9	66.6	-37.7	68.0	-40.3	69.3	-43.7	71.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	33.5	11.8	13.7	20.7	32.3	42.9	51.6	57.2	53.2	45.4	33.9	23.4	15.2
	DBStd	18.55	14.81	13.94	11.07	7.20	6.73	6.28	5.38	5.52	5.89	7.62	11.20	14.58
	HDD50	6502	1184	1016	909	531	240	53	4	29	160	499	799	1078
	HDD65	11489	1649	1436	1374	981	686	403	247	369	589	963	1249	1543
	CDD50	498	0	0	0	0	19	101	228	127	22	1	0	0
	CDD65	10	0	0	0	0	0	2	6	2	0	0	0	0
	CDH74	209	0	0	0	0	5	47	126	31	0	0	0	0
	CDH80	20	0	0	0	0	0	2	17	1	0	0	0	0
Wind	WSAvg	5.3	4.4	4.8	5.9	5.7	6.3	6.4	5.5	5.0	5.6	5.1	4.3	4.3
Precipitation	PrecAvg	24.0	1.5	1.2	1.1	1.3	1.8	2.3	3.7	3.0	2.4	2.0	1.9	1.6
	PrecMax	33.8	2.4	2.2	2.0	2.0	3.5	3.8	6.6	6.7	5.5	4.9	3.9	2.7
	PrecMin	17.1	0.3	0.3	0.4	0.5	0.6	0.4	1.0	1.1	0.6	0.4	0.8	0.3
	PrecStd	4.2	0.6	0.4	0.5	0.5	0.8	0.9	1.3	1.7	1.1	1.1	0.9	0.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	40.1	41.2	44.9	60.9	74.1	79.1	82.6	77.5	66.6	53.6	47.5	41.1
		MCWB	36.6	37.7	38.6	47.0	56.5	59.6	63.7	61.9	56.3	49.0	45.2	38.3
	2%	DB	37.2	38.9	41.3	52.6	68.1	74.2	77.3	73.4	62.0	50.1	42.0	38.0
		MCWB	34.9	35.9	36.5	42.6	53.0	57.5	61.2	60.2	54.5	47.4	40.2	35.8
	5%	DB	33.8	35.4	38.8	48.1	62.8	69.3	73.7	69.3	58.4	47.7	38.9	35.2
		MCWB	32.0	33.1	34.8	40.2	50.6	55.3	60.0	58.8	52.4	45.2	37.5	33.6
	10%	DB	31.2	32.2	36.3	44.3	57.1	65.3	70.2	65.4	55.7	45.3	36.3	33.2
		MCWB	30.0	30.5	33.0	37.9	47.9	53.5	58.9	57.1	50.9	42.8	35.0	32.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	37.0	37.8	39.4	47.6	57.7	62.5	66.9	64.4	58.3	50.3	45.5	39.0
		MCDB	39.5	40.6	43.9	59.5	69.7	73.3	75.4	72.7	64.0	52.3	47.4	40.6
	2%	WB	34.9	36.2	37.1	43.6	54.8	59.2	63.7	61.9	55.7	47.8	40.5	35.9
		MCDB	37.0	38.8	40.7	51.2	64.8	69.7	72.2	69.9	60.0	49.7	41.7	37.6
	5%	WB	32.3	33.2	35.1	40.9	51.7	56.9	62.0	59.9	53.7	45.5	37.6	33.9
		MCDB	33.6	35.1	38.4	47.0	61.9	66.4	70.7	67.1	57.4	47.4	38.6	35.0
	10%	WB	29.8	30.5	33.3	38.6	48.3	54.8	60.1	58.0	51.8	43.1	35.0	32.1
		MCDB	31.0	32.0	36.1	43.6	56.3	63.2	68.0	64.6	54.9	45.0	36.0	33.2
Mean Daily Temperature Range	5% DB	MDBR	15.8	17.6	20.5	17.8	19.3	18.8	18.9	18.9	16.2	12.3	11.1	14.3
		MCDBR	14.3	14.1	17.7	25.0	30.2	26.3	26.5	26.5	21.8	14.0	9.6	13.9
		MCWBR	13.3	12.6	13.9	15.8	17.0	12.4	12.5	15.0	14.9	11.4	9.0	13.1
	5% WB	MCDBR	13.6	13.7	16.3	22.4	28.0	22.8	22.4	22.7	18.4	12.5	9.2	12.5
		MCWBR	12.8	12.4	13.1	14.8	16.5	11.7	11.9	13.5	13.5	10.6	8.6	11.8
Clear-Sky Solar Irradiance	taub	0.188	0.260	0.289	0.323	0.330	0.329	0.341	0.329	0.303	0.285	0.337	N/A	
	taud	1.967	2.199	2.261	2.330	2.480	2.510	2.486	2.509	2.571	2.448	2.744	N/A	
	Ebn at Noon	117	207	253	270	280	282	274	268	254	203	113	N/A	
	Edh at Noon	8	17	26	31	29	29	29	26	20	14	7	N/A	
All-Sky Solar Radiation	RadAvg	42	224	664	1209	1495	1666	1522	1141	652	269	64	16	
	RadStd	3	34	68	97	176	129	170	119	80	43	12	1	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (37 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page